Algebra Monsoon 2026 Lecture 5

Free monoids

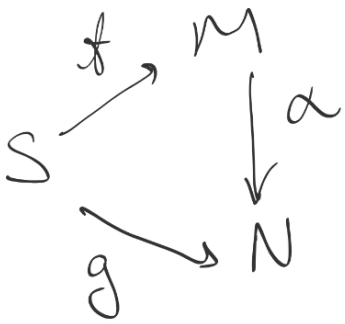
Let S be a set. Consider pairs

$$(M, f)$$

consisting of a monoid M and a mapping $f : S \to M$.

A morphism $\alpha : (M, f) \to (N, g)$ of two such pairs is a ~~mapping~~ morphism of monoids $\alpha : M \to N$ such that $g = f \circ \alpha$.

Equivalently, the taiangle



Commutes.

Clearly, for every pair (M, f),

$$Id_M : (M, f) \longrightarrow (M, f)$$

is a morphism of pairs.

Also, if $\alpha : (M, f) \longrightarrow (N, g)$ and $\beta : (N, g) \longrightarrow (P, h)$ are morphisms of pairs, then so is

$$\beta \circ \alpha : (M, f) \longrightarrow (P, h).$$

Def. A morphism of pairs

$$\alpha : (M, f) \longrightarrow (N, g)$$

is said to be an isomorphism if there exists a morphism of pairs

$$\beta : (N, g) \longrightarrow (M, f)$$

such that

$$\beta \circ \alpha = Id_M$$

and $\alpha \circ \beta = Id_N$.

We say that a pair (L, i) is an initial pair if for every pair (M, f), there is a unique morphism of pairs $(L, i) \longrightarrow (M, f)$.
(universal property of the initial pair (L, i)).

Question: Does such an initial pair exist?

Proposition If (L, i) is an initial pair, the only morphism $\alpha : (L, i) \longrightarrow (L, i)$ is $\alpha = Id_L$.

Proof This is a direct consequence of the definition. □

If an initial pair exists, it is unique up to a unique isomorphism:

Proposition If (L_1, i_1) and (L_2, i_2) are both initial pairs, then the unique morphism of pairs

$$\alpha : (L_1, i_1) \longrightarrow (L_2, i_2)$$

is an isomorphism of pairs.

is an isomorphism.

Proof By the universal property of the initial pair (L_2, i_2), there is a unique morphism of pairs

$$\beta : (L_2, r_2) \longrightarrow (L_1, r_1).$$

By the previous proposition, the morphisms of pairs

$$\beta \circ \alpha : (L_1, r_1) \longrightarrow (L_1, r_1)$$

and $$\alpha \circ \beta : (L_2, r_2) \longrightarrow (L_2, r_2)$$

are equal to Id_{L_1} and Id_{L_2} respectively.

It follows that α is an isomorphism.

Construction of an initial pair:

Let $F_m(S)$ be the set of all sequences

$$(s_1, s_2, \ldots, s_m)$$

where $m \in \mathbb{N}$ and $s_1, \ldots, s_m \in S$, including the empty sequence

$$e = ()$$

Define a law of composition on $L(S)$ by

$$(s_1, \ldots, s_m)(t_1, \ldots, t_n) = (s_1, \ldots, s_m, t_1, \ldots, t_n),$$

where $m, n \in \mathbb{N}$ and $s_1, \ldots, s_m, t_1, \ldots, t_n \in S$.

This makes $F_m(S)$ into a monoid (with the empty sequence ϵ as the neutral element).

We have the injection $i : S \to F_m(S)$,

$$s \longmapsto (s).$$

- The subset $i_*(S)$ generates the monoid $F_m(S)$. (see the corresponding statement for free magmas).

Def The pair $(F_m(S), i)$ is called the **free monoid** on the set S.

Example : • $\mathbb{N}$ is the free monoid on $\{1\}$

• $\mathbb{N}^*$ is the free monoid on the set of prime numbers.

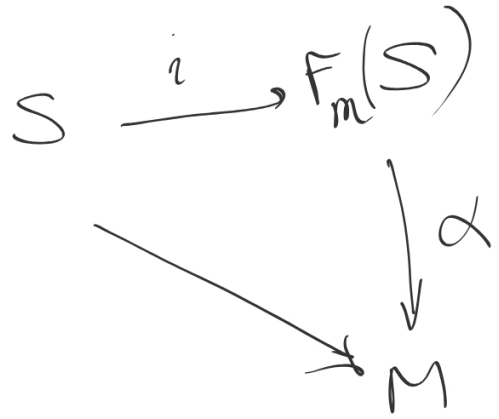
We show that $(F_m(S), i)$ has the universal property justifying its name.

Proposition (Universal property of the free monoid)

For every pair (M, f) consisting of a monoid M and a mapping $f: S \to M$, there is a unique morphism of monoids

$$\alpha : F_m(S) \to M$$

such that $f = \alpha \circ i$ (the triangle



commutes).

Proof For $x = (s_1, \dots, s_m)$ in $F_m(S)$,

define $\alpha(x) = f(s_1) \dots f(s_m)$.

so that $\alpha(e) = e$ (the empty product in a monoid is equal to the neutral element). By construction

$$f = \alpha \circ i .$$

If $y = (t_1, \dots, t_n)$ is another element of $F_m(S)$, then

$$\begin{aligned}\alpha(xy) &= \alpha((s_1, \dots, s_m, t_1, \dots, t_n)) \\ &= f(s_1) \cdots f(s_m) f(t_1) \cdots f(t_n) \\ &= \alpha(x)\alpha(y),\end{aligned}$$

so α is a morphism of monoids.

If $\alpha' : F_m(S) \to M$ is another morphism of monoids such that

$$f = \alpha' \circ i,$$

then α and α' agree on $i_*(S)$.

As $i_*(S)$ generates the monoid $F_m(S)$, we have $\alpha = \alpha'$. □

Proposition For every set S and every monoid M, the mapping

$$\underset{\mathcal{C}_{sets}}{Hom_{\mathcal{E}}}(S, M) \longrightarrow \underset{\mathcal{C}_{monoids}}{Hom_{\mathcal{M}}}(F_{\mathcal{M}}(S), M)$$

sending a mapping $f : S \to M$ to the unique morphism of monoids

$$\alpha : F_{\mathcal{M}}(S) \to M$$

such that $f = \alpha \circ i$ is a bijection.

Proof It is injective because f can be recovered from α as $f = \alpha \circ i$, and it is surjective because every morphism of monoids $\beta : F_{\mathcal{M}}(S) \to M$ comes from the mapping $\beta \circ i : S \to M$.

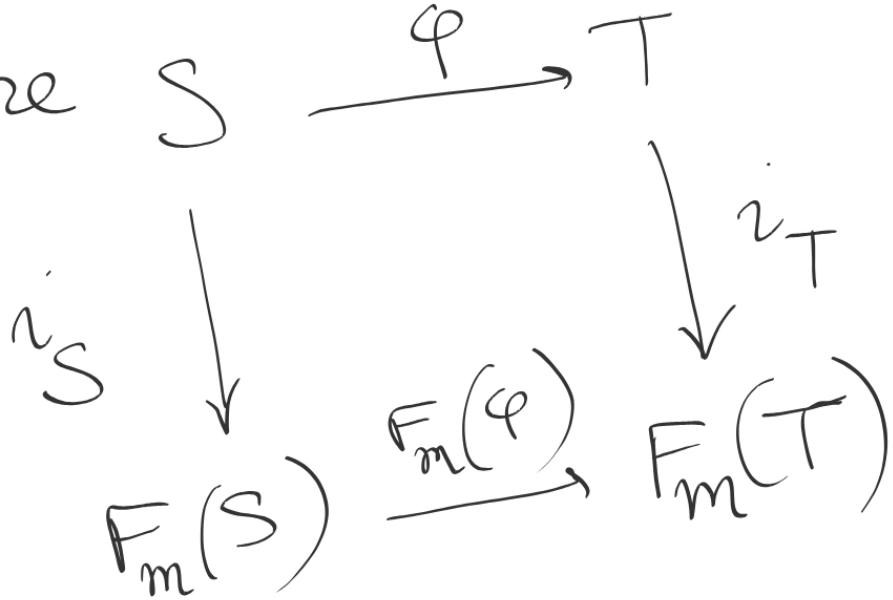
□

Question If we have a mapping of sets $\varphi : S \longrightarrow T$,

do we get a morphism of monoids

$$F_m(\varphi) : F_m(S) \longrightarrow F_m(T).$$

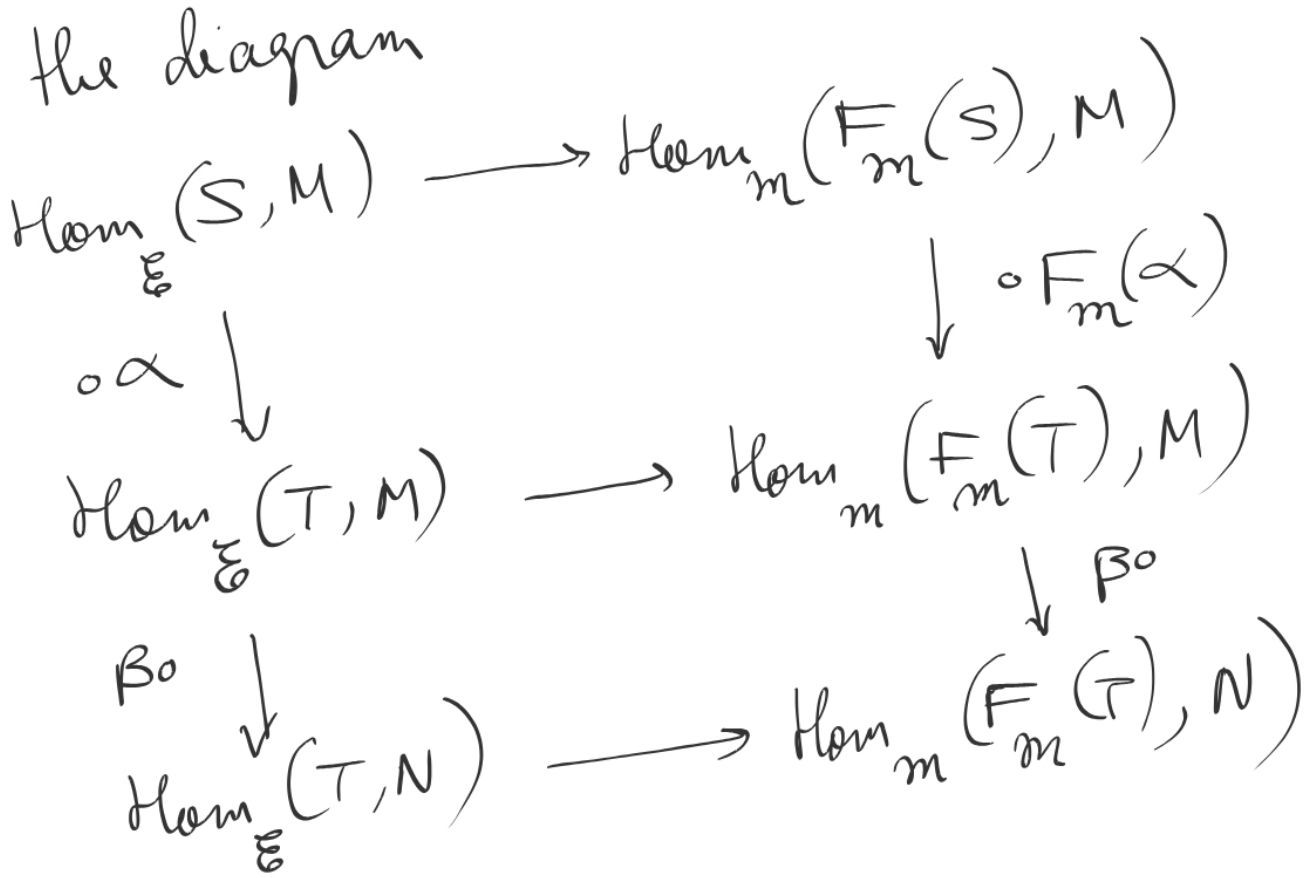
Does the square



commute ?

□

Proposition If $\alpha : T \to S$ is a mapping and $\beta : M \to N$ a morphism of monoids, then the diagram



is commutative.

Exercise Let S be a set and $(F_u(S), j)$ the free unital magma on S.

Consider the smallest congruence C on $F_u(S)$ such that

$$(xy)z \sim_C x(yz)$$

for all $x, y, z \in F_u(S)$.

1) Show that the quotient magma $F_u(S)/C$ is a monoid.

2) Let $\pi : F_u(S) \to F_u(S)/C$ be the canonical surjection. Show that $(F_u(S)/C, \pi \circ j)$ is the free monoid on S.

3) Make the resulting isomorphism of monoids

$$F_u(S)/C \longrightarrow F_m(S)$$

explicit by specifying the composite morphism

$$F_u(S) \longrightarrow F_m(S).$$

of unital magmas.

Free commutative magma

Let S be a set. Let $\mathbb{N}^{(S)}$ be the set of mappings $S \xrightarrow{x} \mathbb{N}$ such that

$$x(s) = 0 \text{ for all but finitely many } s \in S.$$

For $x, y \in \mathbb{N}^{(S)}$, define

$$x + y \in \mathbb{N}^{(S)}$$

by $(x+y)(s) = x(s) + y(s)$ for every $s \in S$. This law of composition makes $\mathbb{N}^{(S)}$ into a commutative monoid, with the constant mapping 0 as the neutral element.

For each $s \in S$, let $e_s \in \mathbb{N}^{(S)}$ be the mapping such that

$$e_s(s) = 1$$
$$e_s(t) = 0 \text{ if } t \neq s.$$

Define $S \longrightarrow \mathbb{N}^{(S)}$ by

$$i(s) = e_s \text{ for all } s \in S.$$

Note that, for every $x \in \mathbb{N}^{(S)}$, we have

$$x = \sum_{s \in S} x(s)\, e_s .$$

(The monoid $\mathbb{N}^{(S)}$ is generated by the subset $i_*(S)$)

so that if $\sum_{s \in S} x(s) e_s = \sum_{s \in S} y(s) e_s$

for some $x, y \in \mathbb{N}^{(S)}$, then $x = y$.

Proposition $(\mathbb{N}^{(S)}, i)$ is the free commutative monoid on S.

In other words, for any commutative monoid M, the mapping

$$\underset{\mathcal{C}}{\mathrm{Hom}}(F(S), M) \longrightarrow \underset{\mathcal{E}}{\mathrm{Hom}}(S, M)$$

($\mathcal{C}$: commutative monoids; $\mathcal{E}$: sets)

$$\varphi \longmapsto \varphi \circ i$$

is a bijection.

Proof Given $f \in \underset{\mathcal{E}}{\mathrm{Hom}}(S, M)$,

take $\varphi \in \underset{\mathcal{C}}{\mathrm{Hom}}(F(S), M)$

to be

$$\varphi\left(\sum_{s \in S} x(s)\, e_s\right) = \prod_{s \in S} f(s)^{x(s)}$$

□

Exercise Let S be a set, $F_m(S)$ the free monoid on S, and C the smallest congruence on $F_m(S)$ such that $xy \sim_C yx$ for all $x, y \in F_m(S)$.

Show that the quotient monoid $F_m(S)/C$ is a free commutative monoid on S. Make the resulting isomorphism

$$F_m(S)/C \longrightarrow \mathbb{N}^{(S)}$$

explicit by specifying the composite morphism $F_m(S) \longrightarrow \mathbb{N}^{(S)}$.